MILITARY STANDARD

CARLOADING

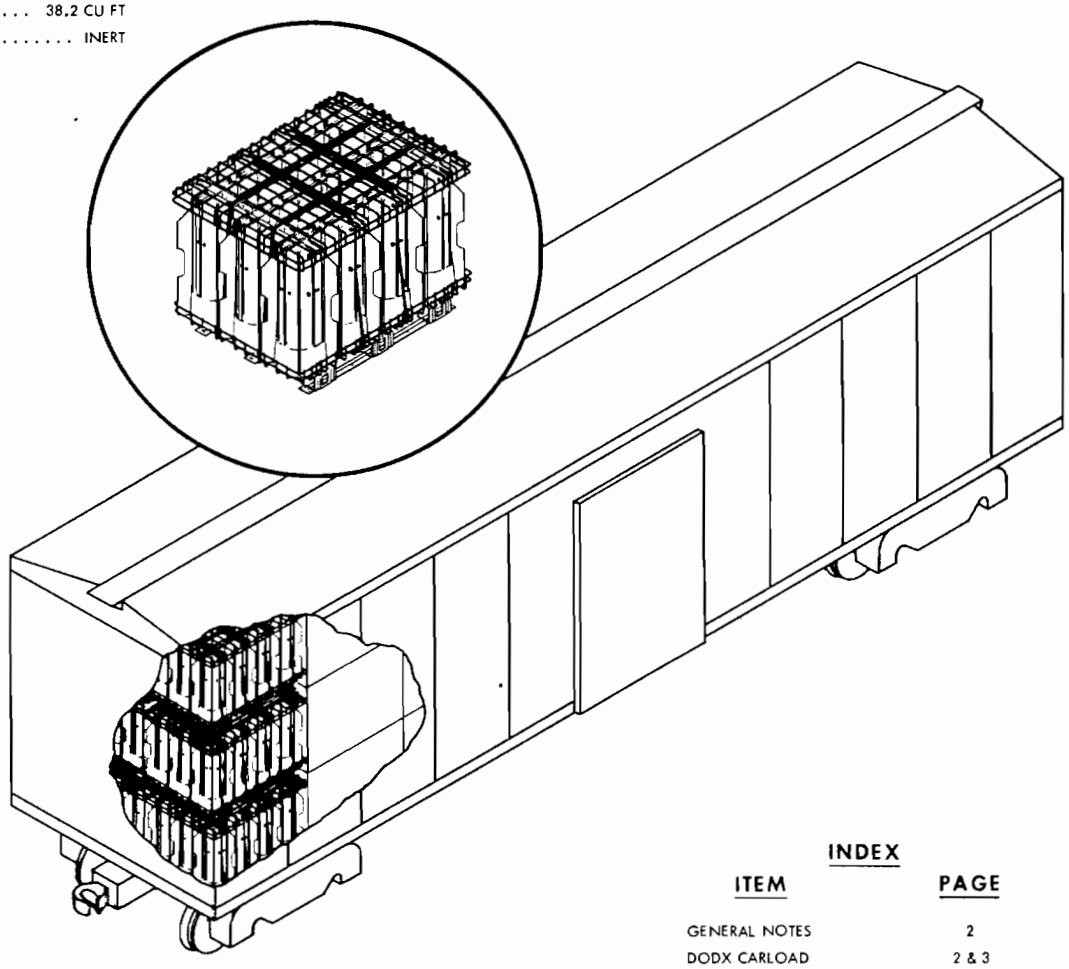
MIL-STD-1325-166
(NAVY)

17 JULY 1973

FIN ASSEMBLY, BOMB, MK 15 AND MODS (SNAKEYE) IN PALLET ADAPTER ADU-383/E FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/242
WEIGHT . . . . . 971 LBS
DIMENSIONS . . . 48 1/4 L x 36 1/4 W x 37 3/4 H
CUBE . . . . . 38.2 CU FT
DOT HAZARD CLASSIFICATION . . . . . INERT



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DODX CARLOAD	2 & 3
40'-6" CARLOAD, COML	4 & 5
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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236(NAVY).

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Gray NWHL 7/10/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 7/11/73
SIGNATURE (AIR) SYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES NOT APPLICABLE

MILITARY ASSISTANT
SIGNATURE DATE

ORIGINATOR Charles McBride 7/10/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

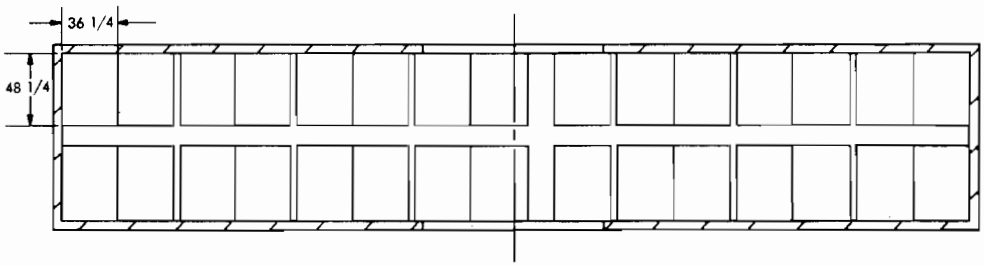
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF MK 15 AND MODS BOMB FIN ASSEMBLY (SKAKEYE) PALLETIZED IN ACCORDANCE WITH WR-54/242.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, AND CLOSE AND SEAL BOXCAR DOORS.

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 90 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
5	LATERAL FLOOR MEMBER	2 x 6 x 107	2	CAR FLOOR	2 PER FOOT	16d
4	CROSS BRACE	2 x 6 x W*	90	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 37	60	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	60	SEE 3	-	-
1	SLEEPER	2 x 4 x 36	30	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** = PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

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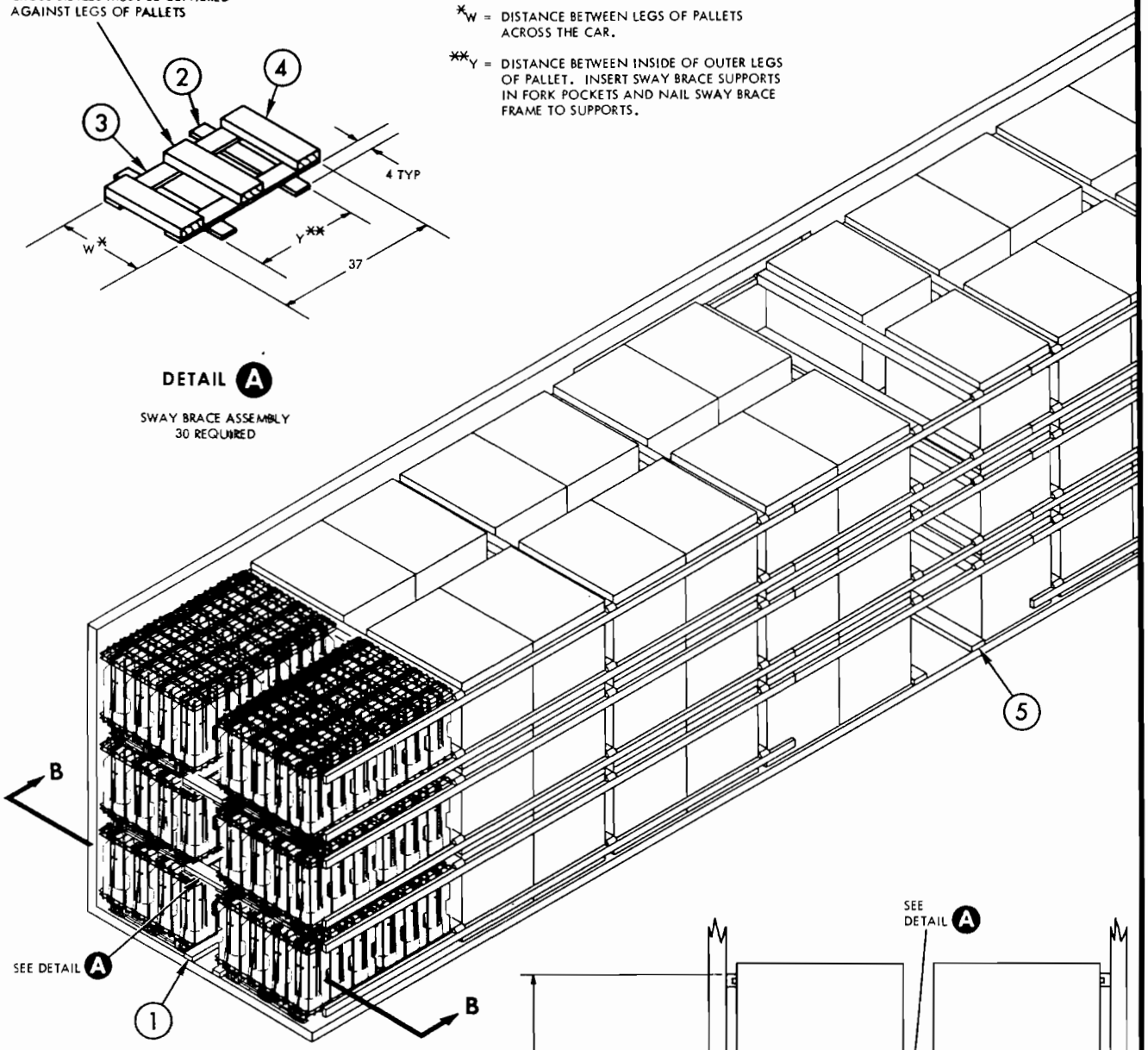
CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS

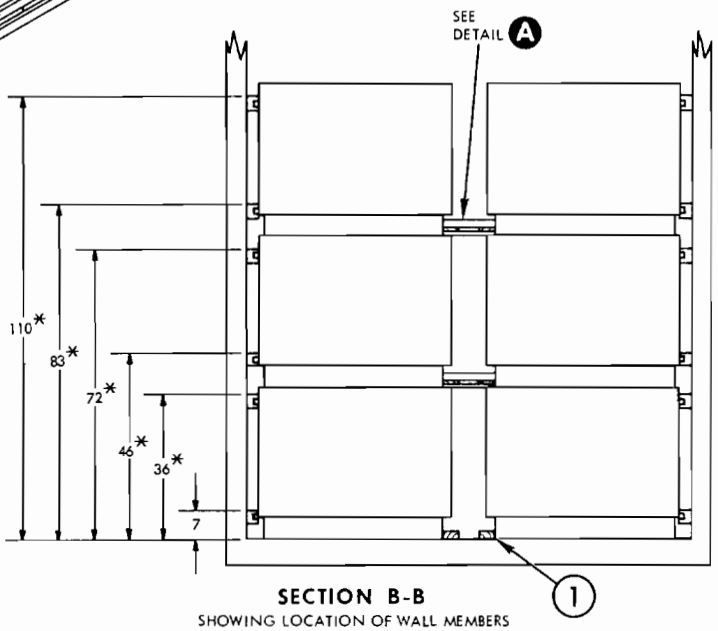
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

SWAY BRACE ASSEMBLY
30 REQUIRED





SECTION B-B

SHOWING LOCATION OF WALL MEMBERS USED FOR CROSSMEMBERS

*DOORWAY MEMBER LOCATIONS

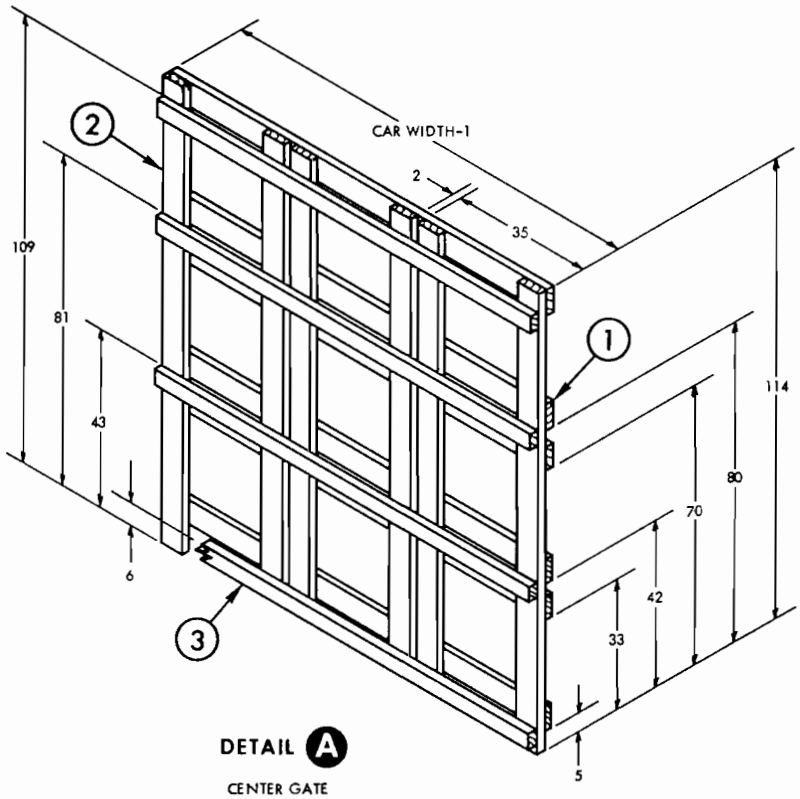
CARLOAD DATA

NUMBER OF UNIT LOADS	90
NUMBER OF CROSSMEMBERS REQ'D	46
LOAD WEIGHT	87,390 LBS.
DUNNAGE WEIGHT	845 LBS
CARLOAD WEIGHT	88,235 LBS.

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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 81 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.



DETAIL A

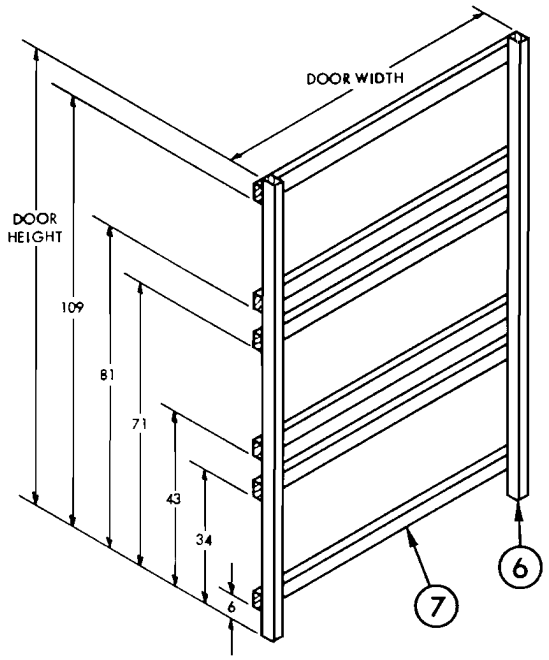
CENTER GATE
2 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
7	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	6	2 PER JOINT	10d
6	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
5	GATE HOLD DOWN	2 x 4 x 60	2	3	3 PER JOINT	10d
4	STRUT	4 x 4 x WEDGE FIT *	24	2	2 PER JOINT	16d
3	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	2	3 PER JOINT	10d
2	CENTER GATE VERTICAL	2 x 6 x 114	12	SEE 1	-	-
1	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	2	3 PER JOINT	10d

LIST OF MATERIALS AND NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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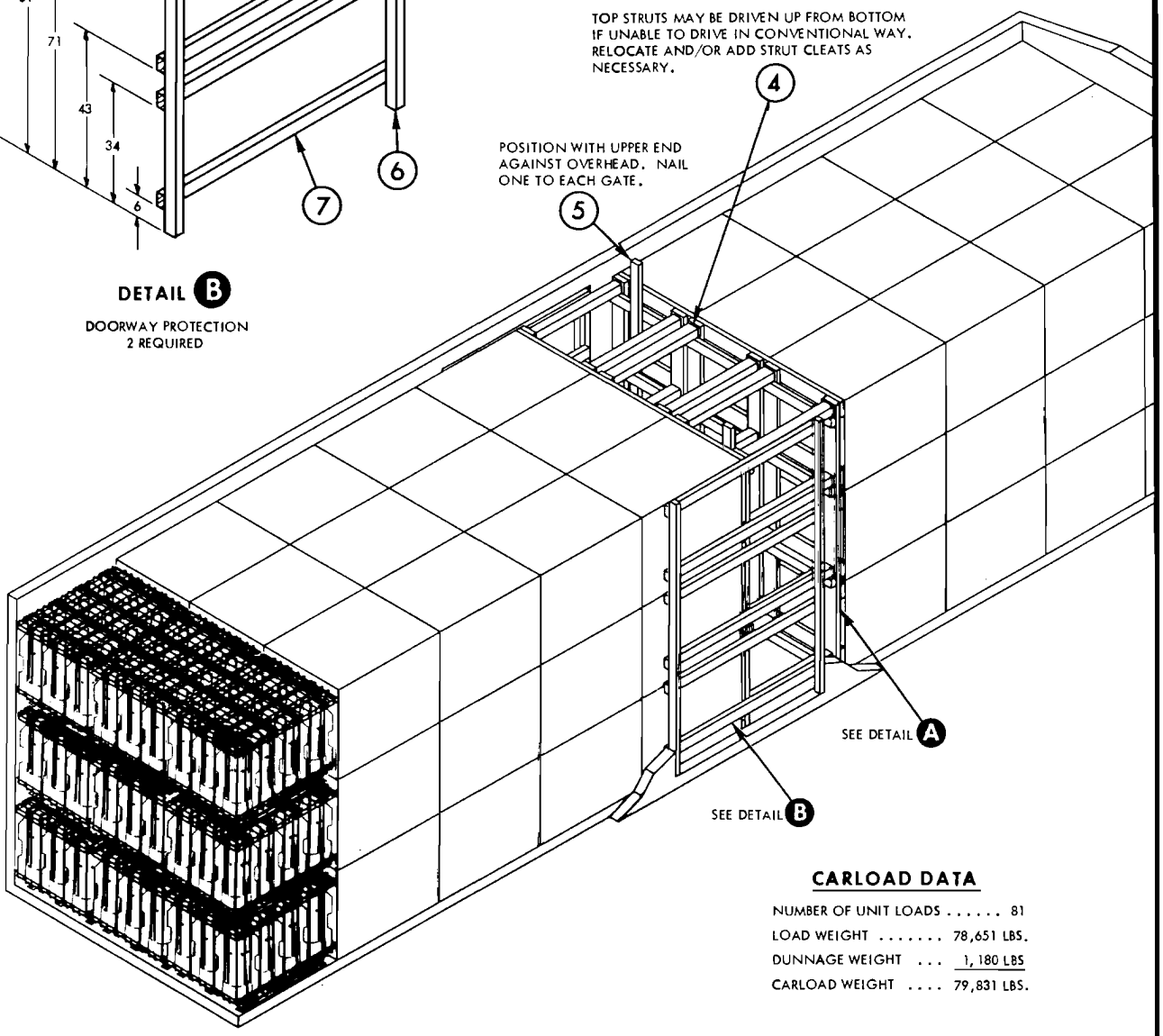
DETAIL B

DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

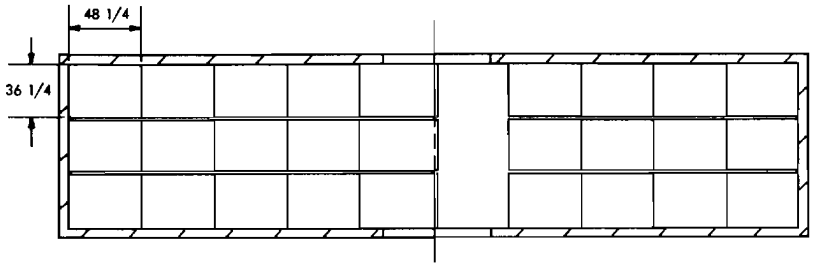
NOTE:

TOP STRUTS MAY BE DRIVEN UP FROM BOTTOM IF UNABLE TO DRIVE IN CONVENTIONAL WAY. RELOCATE AND/OR ADD STRUT CLEATS AS NECESSARY.



CARLOAD DATA

NUMBER OF UNIT LOADS	81
LOAD WEIGHT	78,651 LBS.
DUNNAGE WEIGHT	1,180 LBS
CARLOAD WEIGHT	79,831 LBS.

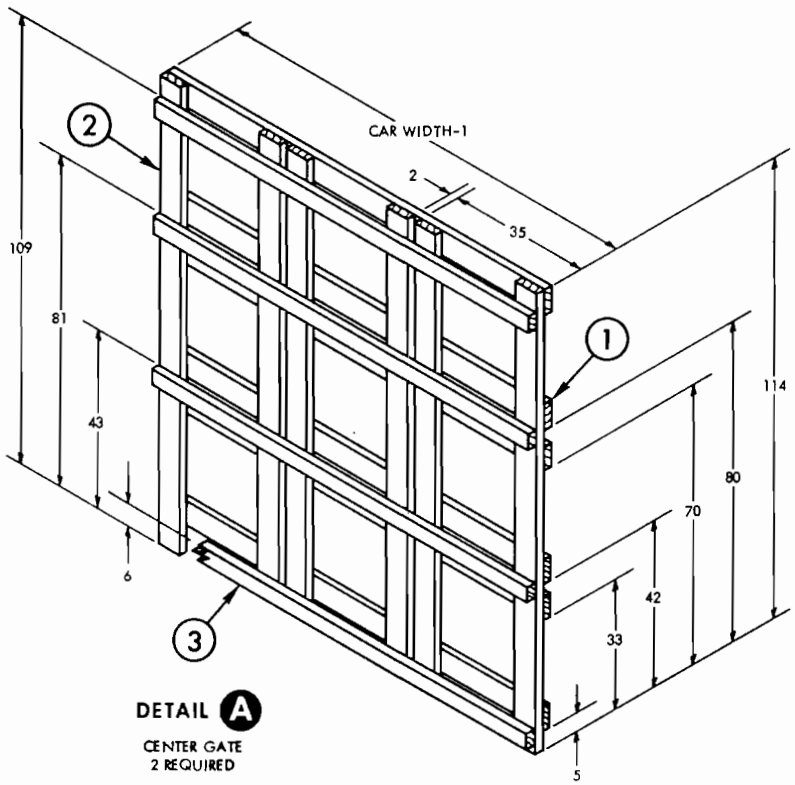


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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 108 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH , SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION, PIECES 6 AND 7 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



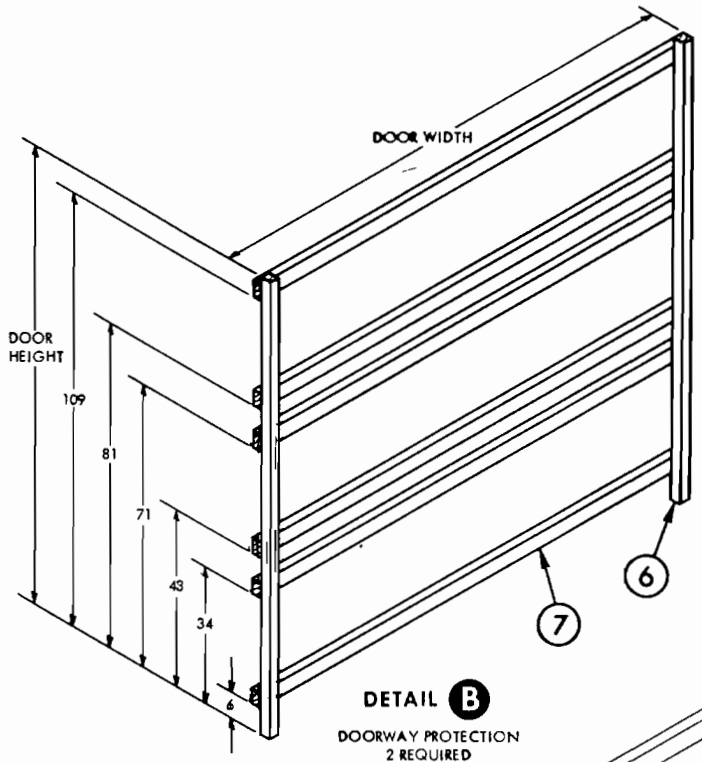
DETAIL A
CENTER GATE
2 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
7	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	6	2 PER JOINT	10d
6	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
5	GATE HOLD DOWN	2 x 4 x 60	2	3	3 PER JOINT	10d
4	STRUT	4 x 4 x WEDGE FIT*	24	2	2 PER JOINT	16d
3	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	2	3 PER JOINT	10d
2	CENTER GATE VERTICAL	2 x 6 x 114	12	SEE 1	-	-
1	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	2	3 PER JOINT	10d

LIST OF MATERIALS AND NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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DETAIL B

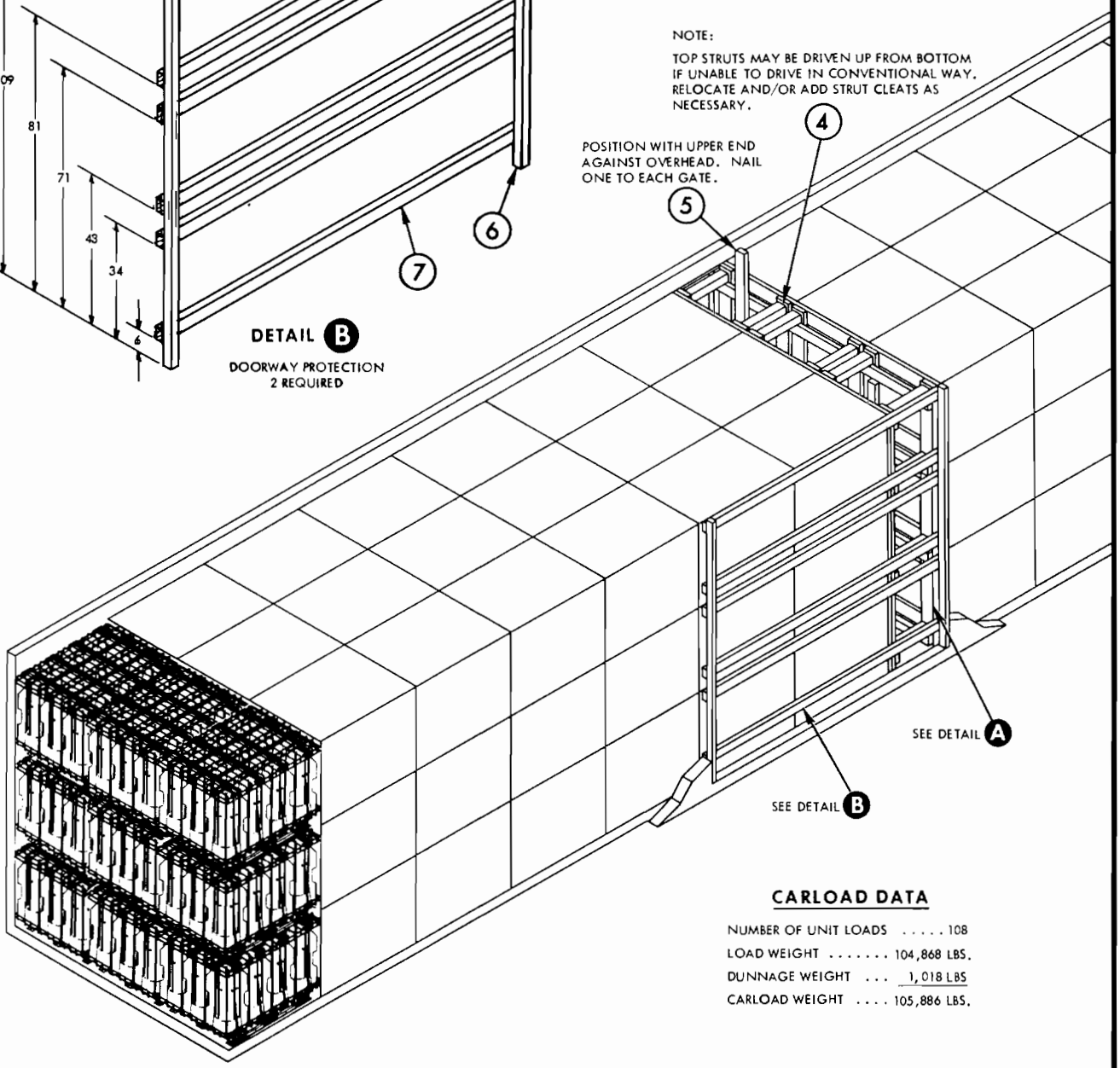
DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

NOTE:

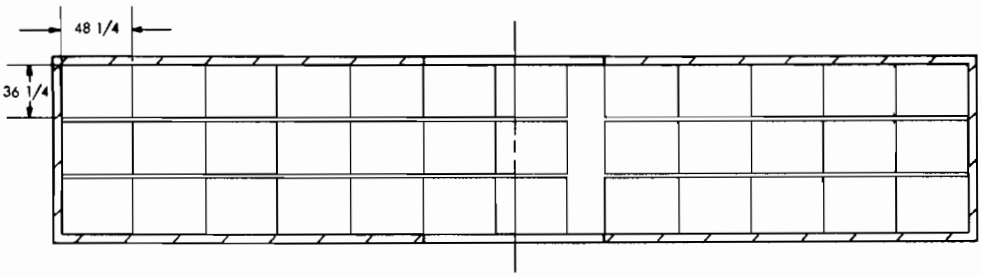
TOP STRUTS MAY BE DRIVEN UP FROM BOTTOM IF UNABLE TO DRIVE IN CONVENTIONAL WAY. RELOCATE AND/OR ADD STRUT CLEATS AS NECESSARY.

POSITION WITH UPPER END AGAINST OVERHEAD. NAIL ONE TO EACH GATE.



CARLOAD DATA

NUMBER OF UNIT LOADS	108
LOAD WEIGHT	104,868 LBS.
DUNNAGE WEIGHT	1,018 LBS
CARLOAD WEIGHT	105,886 LBS.



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